SOLUTIONS

Module:	Fundamentals of DSP		
Module Code	QHE5107	Paper	A
Time allowed	3 hrs	Filename	"QHE5107-Fundamentals of DSP - First Sit - Draft version - Without Answers"
Rubric	ANSWER ALL FOUR QUESTIONS		
Examiners	Dr Fabrício de Oliveira Ourique	Dr Jesús Requena Carrion	

北京邮电大学 | Queen Mary University of London

QHE5107 A

For examiners' use only	
1	
2	
3	
4	
Total	

Queen Mary School Hainan 2024/25

QHE5107 Fundamentals of DSP

Paper A

Time allowed 3 hours

Answer ALL questions

Complete the information below yourself very carefully.

QM student number

BUPT student number

Class number

NOT allowed: Any Electronic devices including calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must not take answer books, used or unused, from the examination room.**
2. Write only in black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**
4. If you use Supplementary Answer Book, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Fabrício de Oliveira Ourique, Dr Jesús Requena Carrion

Filename: "QHE5107-Fundamentals of DSP - First Sit - Draft version - Without Answers" No answerbook required

Question 1 **[25 marks]**

Consider the discrete-time system shown in Fig.1.

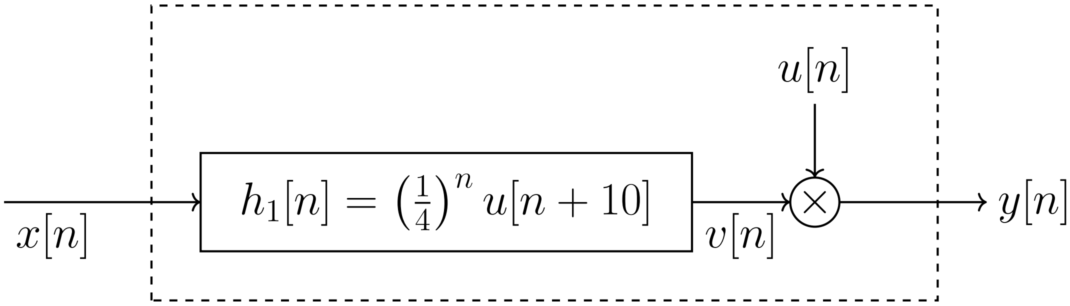


Figure 1: Discrete-Time System

(a) What is the expression for the output $y[n]$ as a function of the input signal $x[n]$? **(5 marks)**

Solution:

$$v[n] = x[n] \star h_1[n] = x[n] \star \left(\frac{1}{4}\right)^n u[n+10] \text{ and} \qquad \text{(Basic: 2 marks)}$$

$$y[n] = u[n]v[n] = u[n]\left(x[n] \star \left(\frac{1}{4}\right)^n u[n+10]\right) \qquad \text{(Basic: 3 marks)}$$

(b) Compute the impulse response of the system. **(5 marks)**

Solution:
The impulse response, $h[n]$, is

$$h[n] = \mathcal{T}\{\delta[n]\} = u[n]\left(\delta[n] \star \left(\frac{1}{4}\right)^n u[n+10]\right) \qquad \text{(Basic: 1 mark)}$$

$$= u[n]\left(\left(\frac{1}{4}\right)^n u[n+10]\right) \qquad \text{(Medium: 2 marks)}$$

$$h[n] = \left(\frac{1}{4}\right)^n u[n] \qquad \text{(Advanced: 2 marks)}$$

(c) Is the system Linear Time-Invariant (LTI)? **(5 marks)**

Solution:
To be Time-Invariant, we need

$$y[n-n_0] = \mathcal{T}\{x[n-n_0]\} \qquad \text{(Basic: 1 mark)}$$

As

$$\mathcal{T}\{x[n-n_0]\} = u[n]\left(x[n-n_0] \star \left(\frac{1}{4}\right)^n u[n+10]\right) \qquad \text{(Basic: 2 marks)}$$

is not equal to

$$y[n-n_0] = u[n-n_0]\left(x[n-n_0] \star \left(\frac{1}{4}\right)^{(n-n_0)} u[n+10-n_0]\right) \qquad \text{(Medium: 2 marks)}$$

the system is not LTI.

(d) Is the system causal? **(5 marks)**

Solution:
To be causal, we need

$$h[n] = 0, \forall n < 0$$ (Basic: 2 marks)

The impulse response, $h[n]$, is

$$\begin{aligned} h[n] = \mathcal{T}\{\delta[n]\} &= u[n]\left(\delta[n] \star \left(\frac{1}{4}\right)^n u[n+10]\right) \\ &= u[n]\left(\left(\frac{1}{4}\right)^n u[n+10]\right) \\ h[n] &= \left(\frac{1}{4}\right)^n u[n] \end{aligned}$$

The system is causal. (Medium: 3 marks)

(e) Is the the system stable? **(5 marks)**

Solution:
To be stable, we need

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$ (Basic: 1 mark)

As

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=-\infty}^{\infty} \left(\frac{1}{4}\right)^n u[n]$$ (Advanced: 2 marks)

$$= \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n = \frac{4}{3}$$ (Advanced: 1 mark)

The system is stable. (Medium: 1 mark)

Question 2 **[25 marks]**

One of the main problems when collecting electrocardiogram (ECG) signals is the undesired interference of 60 Hz in the sampled signal. This interference is caused by factors such as electrical power, magnetic induction, currents at the electrodes in contact with the patient's skin, and connections between devices like data loggers. Assume that the bandwidth of the desired signal is 1 kHz, i.e.:

$$X_a(f) = 0, |f| > 1\ kHz$$

The continuous signal is converted into a discrete signal using an ideal ADC (Analog to Discrete Converter) operating at a sampling frequency. The resulting discrete signal $x[n] = x_a(nT)$ is then processed by a discrete-time system described by the following difference equation:

$$y[n] = x[n] + ax[n-1] + bx[n-2]$$

The filtered signal $y[n]$ is then converted back to a continuous signal using an ideal DAC (Discrete to Analog Converter).

(a) What is the minimum sampling frequency required to prevent aliasing? **(5 marks)**

Solution:

$f_M = 1kHz$ Bandwidth of the desired signal (Basic: 2 marks)

$f_S > 2kHz$ (Basic: 3 marks)

(b) What type of filter should be used to minimise the interference signal, and why? **(5 marks)**

Solution:
Band Stop (Basic: 2 marks)
Because the interference signal is concentrated in a specific band of the spectrum.
. (Medium: 3 marks)

(c) Design the filter (discrete-time system) to remove the 60 Hz interference by selecting values for f_S, b_1 and b_2 of the difference equation $y[n] = x[n] + b_1 x[n-1] + b_2 x[n-2]$ so that the interference signal, $i_a(t)$, of 60 Hz

$$i_a(t) = A\sin(120\pi t)$$

does not appear in the output signal at the DAC output. You may leave the values for b_1 and b_2 as functions of f_S. **(15 marks)**

Solution:
Selecting the sampling frequency equals to $2kHz$, $f_S = 2kHz$. (Basic: 2 marks)

The system function is

$$H(z) = 1 + az^{-1} + bz^{-2}$$ (Medium: 3 marks)

The system has two zeros. By placing the zeros at the digital frequency equivalent to $60Hz$, the interference noise will be removed. (Medium: 3 marks)

The digital frequency equivalent to $60Hz$, ω_i, is

$$\omega_i = \frac{2\pi 60}{f_S}$$ (Advanced: 2 marks)

For the system to have two zeros at the desired position, we need

$$H(z) = \left(1 - e^{-j\omega_i} z^{-1}\right)\left(1 - e^{j\omega_i} z^{-1}\right) =$$ (Advanced: 3 marks)

$$H(z) = 1 - 2\cos(\omega_i) z^{-1} + z^{-2}$$ (Medium: 2 marks)

Therefore,

$$f_S = 2kHz$$

$$a = -2\cos(\omega_i) = -2\cos\left(\frac{120\pi}{f_S}\right)$$

$$b = 1$$

Question 3 **[25 marks]**

For the system defined below:

$$y[n] = x[n] - 0.25y[n-2]$$

(a) Classify the system as recursive or non-recursive. **(5 marks)**

Solution:
Since $y[n]$ depends on $y[n-2]$, the system is recursive, (Basic: 3 marks)
as it uses a past output value in determining the current output. (Basic: 2 marks)

(b) Obtain the system function, $H(z)$, and classify the system as FIR or IIR **(5 marks)**

Solution:
The system function is

$$H(z) = \frac{1}{1+0.25z^{-2}} = \frac{1}{(1-j0.5z^{-1})(1+j0.5z^{-1})} \qquad \text{(Basic: 2 marks)}$$
$$= \frac{A_1}{(1-j0.5z^{-1})} + \frac{A_2 = A_1^*}{(1+j0.5z^{-1})} \qquad \text{(Medium: 1 mark)}$$

Appling the inverser z-Transform

$$h[n] = 2|A|(0.5)^n \cos\left(\frac{\pi}{2}n + \theta\right) u[n], \text{ where } \theta \text{ is the phase of } A. \qquad \text{(Advanced: 1 mark)}$$

The system is IIR (Medium: 1 mark)

(c) Classify the system as a Low-Pass filter, High-Pass filter, Band-Pass filter, or Band-Stop filter. **(5 marks)**

Solution:
The system function is

$$H(z) = \frac{1}{1+0.25z^{-2}} \qquad (1)$$

The system has two poles at $\pm j0.5$. (Medium: 2 marks)
Based on the location of the poles, the system is band-pass. (Advanced: 3 marks)

(d) The signal $x[n] = \cos\left(\frac{\pi}{2}n\right)$ is applied to the system's input. What is the output signal $y[n]$? **(10 marks)**

Solution:
The system function is

$$H(z) = \frac{1}{1+0.25z^{-2}}$$

The resquency response is

$$H\left(e^{j\omega}\right) = H(z)|_{z=e^{j\omega}} \quad \text{(Basic: 2 marks)}$$

$$= \frac{1}{1+0.25e^{-j2\omega}} \quad \text{(Basic: 2 marks)}$$

$$H\left(e^{j\omega}\right)\big|_{w=\pi/2} = \frac{1}{1+0.25e^{-j\pi}} = 1.333 \quad \text{(Medium: 3 marks)}$$

The output signal is

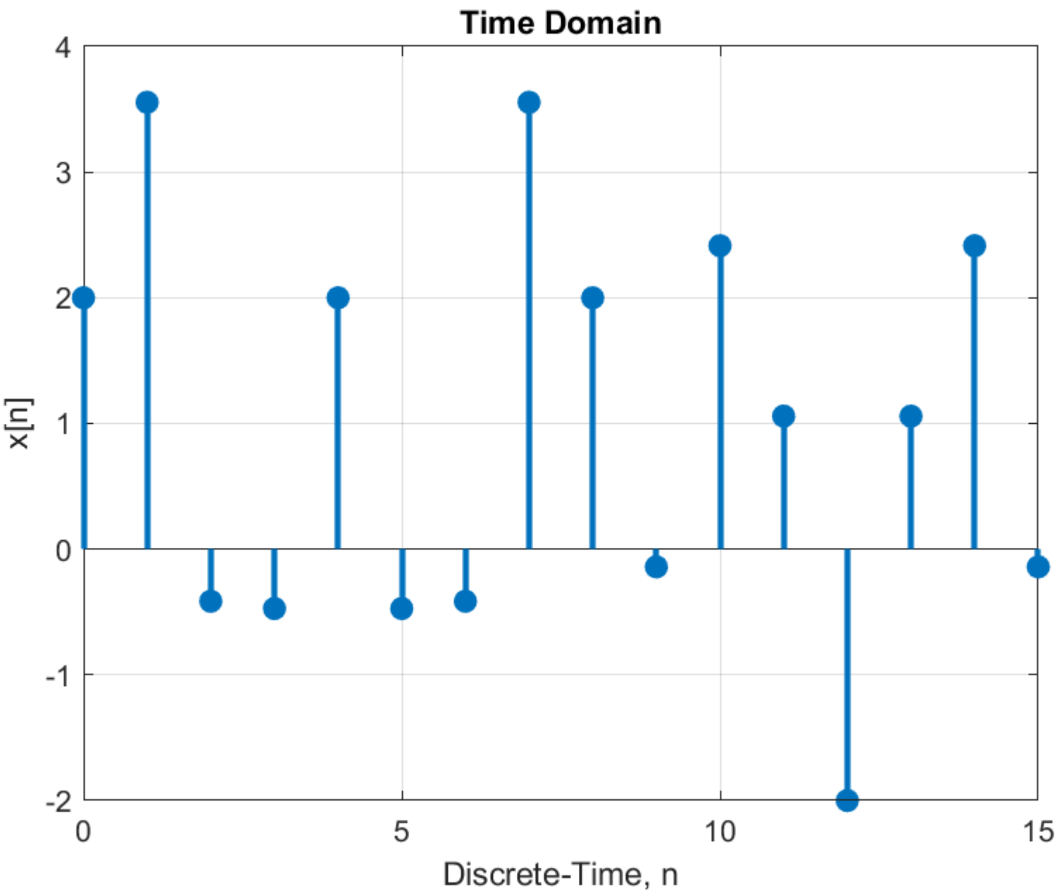
$$y[n] = 1.333\cos\left(\frac{\pi}{2}n\right) \quad \text{(Advanced: 2 marks)}$$

Question 4 **[25 marks]**

A continuous-time signal has a bandwidth of 4 kHz, i.e.,

$$X_c(f) = 0, |f| > 4\ kHz$$

The continuous-time signal is converted into a discrete signal using an ideal ADC (Analog to Discrete Converter) operating at a sampling frequency $f_S = 8\ kHz$, resulting in the following samples of the discrete-time signal $x[n]$, as shown in Fig. 2



n	$x[n]$
0	2.0000
1	3.5549
2	-0.4142
3	-0.4725
4	2.0000
5	-0.4725
6	-0.4142
7	3.5549
8	2.0000
9	-0.1407
10	2.4142
11	1.0583
12	-2.0000
13	1.0583
14	2.4142
15	-0.1407

Figure 2: Discrete-Time signal, $x[n]$, in the Time Domain

The Discrete Fourier Transform (DFT) of the signal is shown in Fig. 3 and Fig. 4, representing the magnitude and phase, respectively.

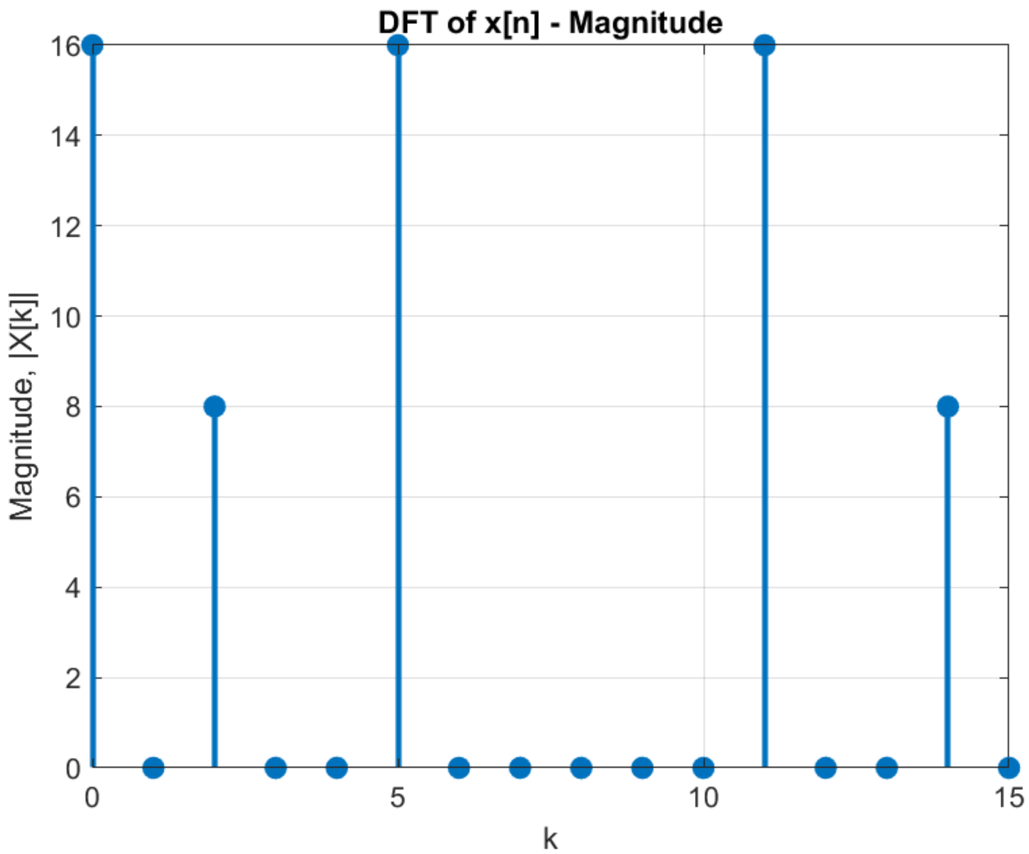


Figure 3: Magnitude of $X[k]$

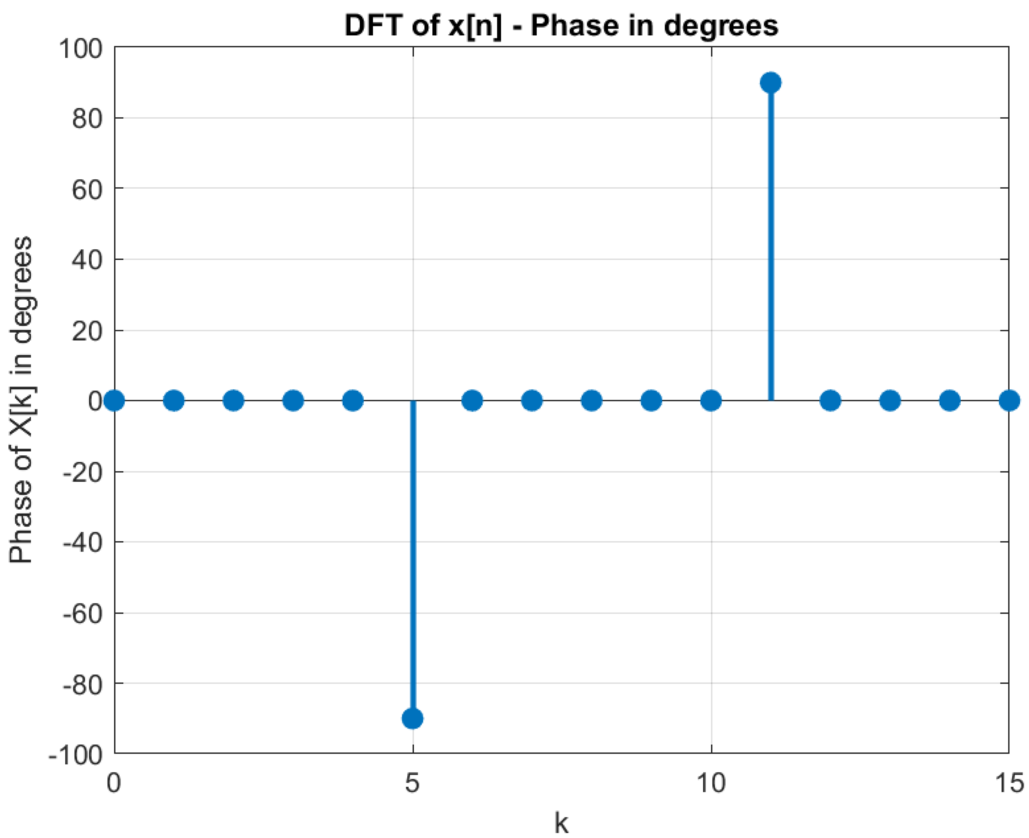


Figure 4: Phase of $X[k]$

(a) What is the DC value of the signal, i.e., the average of the signal? **(5 marks)**

Solution:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk} \qquad \text{(Basic: 1 mark)}$$

$$X[0] = \sum_{n=0}^{N-1} x[n] \qquad \text{(Basic: 2 marks)}$$

$$\frac{1}{N}\sum_{n=0}^{N-1} x[n] = \frac{X[0]}{N} = 1 \qquad \text{(Advanced: 2 marks)}$$

(b) What frequency components does the continuous-time signal contain, in Hertz? **(10 marks)**

Solution:

$$k = 0,\ \omega = 0 \times \frac{2\pi}{16} rad \qquad \text{(Basic: 1 mark)}$$

$$\textbf{equivalent to } 0Hz(DC) \qquad \text{(Medium: 1 mark)}$$

$$k = 2,\ \omega = 2 \times \frac{2\pi}{16} rad \qquad \text{(Basic: 2 marks)}$$

$$\textbf{equivalent to } 1kHz \qquad \text{(Medium: 2 marks)}$$

$$k = 5,\ \omega = 5 \times \frac{2\pi}{16} rad \qquad \text{(Basic: 2 marks)}$$

$$\textbf{equivalent to } 2.5kHz \qquad \text{(Meduim: 2 marks)}$$

(c) What is the expression for $x[n]$? i.e., the inverse discrete Fourier transform of $X[k]$. **(10 marks)**

Solution:
The IDFT is

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}nk}, \quad n = 0, 1, \cdots, N-1 \qquad \text{(Basic: 2 marks)}$$

From Figs. 3 and 4, we have

$$X[k] = 16\delta[k] + 8\delta[k-2] - j16\delta[k-5] + j16\delta[k-11] + 8\delta[k-14] \qquad \text{(Medium: 3 marks)}$$

It follows that

$$
\begin{aligned}
x[n] &= \frac{1}{16}\left(16 + 8e^{j\frac{2\pi}{N}n2} - j16e^{j\frac{2\pi}{N}n5} + j16e^{j\frac{2\pi}{N}n11} + 8e^{j\frac{2\pi}{N}n14}\right) && \text{(Advanced: 2 marks)} \\
&= \frac{1}{16}\left(16 + 8e^{j\frac{2\pi}{N}2n} + 8e^{-j\frac{2\pi}{N}2n} + 16e^{j(\frac{2\pi}{N}5n-\frac{\pi}{2})} + 16e^{-j(\frac{2\pi}{N}5n-\frac{\pi}{2})}\right) \\
x[n] &= 1 + \cos\left(\frac{2\pi}{16}2n\right) + 2\cos\left(\frac{2\pi}{16}5n - \frac{\pi}{2}\right) && \text{(Advanced: 3 marks)} \\
x[n] &= 1 + \cos\left(\frac{2\pi}{16}2n\right) + 2\sin\left(\frac{2\pi}{16}5n\right)
\end{aligned}
$$